Concordance Validation of GenomeGuard, a Lightweight VCF-Based Pharmacogenomic Interpretation Engine, Against PharmCAT Using 1000 Genomes South Asian Whole-Genome Sequencing Data

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Abstract

Background: Pharmacogenomic (PGx) testing holds transformative potential for precision medicine in South Asian populations, where clinically actionable allele frequencies diverge substantially from European-centric reference panels. However, existing PGx interpretation tools are computationally heavy and often require Dockerized Java environments, limiting point-of-care deployment.

Methods: We developed GenomeGuard, a lightweight, Python-native VCF-based PGx interpretation engine that performs rsID-driven star-allele calling, diplotype construction, and CPIC activity-score-based phenotype inference without external dependencies. We conducted a retrospective concordance study benchmarking GenomeGuard against PharmCAT (Pharmacogenomics Clinical Annotation Tool, PharmGKB) using high-coverage (30x) whole-genome sequencing data from 601 South Asian individuals across five 1000 Genomes Project populations (BEB, GIH, ITU, PJL, STU).

Results: Of 13 Tier 1 pharmacogenes amenable to direct head-to-head comparison, all 13 were successfully evaluated. Three additional genes (CYP1A2, CYP2C8, NAT2) were assessed only via internal synthetic unit tests and were not included in concordance statistics because PharmCAT's current allele-definition files do not produce callable results for these genes from short-read VCF data. Across the 13 compared genes and 7,810 sample-gene pairs, GenomeGuard achieved 99.95% diplotype concordance and 99.96% phenotype concordance with PharmCAT. Only 6 sample-level mismatches were observed (4 attributable to allele-definition version differences and 2 to notation divergence). Zero logic-level disagreements were detected. Under the tested deployment configuration, GenomeGuard demonstrated substantially lower execution time, completing the full 601-sample analysis in 2.3 seconds (wall-clock) compared to 3199 seconds (53.3 minutes) for PharmCAT via Docker (1391x difference).

Conclusions: GenomeGuard produces clinically equivalent pharmacogenomic calls to the PharmCAT tool while offering dramatically faster execution suitable for real-time clinical decision support, particularly in resource-constrained South Asian healthcare settings.

Keywords: *pharmacogenomics, concordance study, VCF, South Asian, CPIC, PharmCAT, GenomeGuard, 1000 Genomes, precision medicine, star-allele calling*

1. Introduction

Pharmacogenomics (PGx) translates germline genetic variation into actionable prescribing guidance. The Clinical Pharmacogenetics Implementation Consortium (CPIC) publishes evidence-based guidelines that map star-allele diplotypes to metabolizer phenotypes and, in turn, to drug-specific dosing recommendations [1,2]. Robust computational tools that faithfully implement these guidelines are essential for clinical adoption of PGx testing.

PharmCAT (Pharmacogenomics Clinical Annotation Tool), developed by PharmGKB at Stanford University, is a widely cited open-source PGx interpretation tool [3]. PharmCAT operates as a multi-stage Java pipeline typically deployed via Docker, comprising a VCF preprocessor, Named Allele Matcher, Phenotyper, and Reporter. While highly accurate, its reliance on a Java runtime and Docker containerization introduces computational overhead that may limit deployment in resource-constrained or real-time clinical environments.

South Asian populations are markedly underrepresented in PGx validation studies despite harboring clinically significant allele frequency distributions that differ from European-centric reference panels [4,5]. For example, CYP2C19 poor-metabolizer alleles (affecting clopidogrel response) and CYP3A5 expresser alleles (affecting tacrolimus dosing) occur at substantially different frequencies in South Asian populations compared to European cohorts. Validation against ethnically matched reference data is therefore a prerequisite for responsible clinical deployment.

We developed GenomeGuard, a lightweight Python-native PGx interpretation engine designed for deployment in Indian healthcare settings. GenomeGuard processes standard VCF files through a three-stage pipeline: (i) rsID-based variant identification against a curated GRCh38 coordinate map; (ii) star-allele assignment and diplotype construction using CPIC allele-definition tables; and (iii) phenotype inference via CPIC activity-score logic with gene-specific heuristics for non-CYP pharmacogenes (e.g., VKORC1 sensitivity, NAT2 acetylator status, IFNL3 response genotype). The engine operates without Docker, Java, or network dependencies, enabling sub-second per-sample analysis.

This study presents a systematic concordance validation of GenomeGuard against PharmCAT using 601 South Asian whole-genome sequences from the 1000 Genomes Project, reporting per-gene diplotype and phenotype concordance, detailed mismatch triage, execution time benchmarks, and population-stratified results. We additionally provide supplementary internal consistency checks for all remaining genes in the 20-gene panel.

2. Methods

2.1 Study Design

A retrospective concordance study comparing two independent PGx interpretation pipelines (GenomeGuard v1.0.0 and PharmCAT v2.15.4 via Docker image pgkb/pharmcat:2.15.4) run on identical input VCF files derived from the same source cohort. Both tools were executed on the same hardware (Windows 11 workstation, 16 GB RAM) to ensure comparability of timing measurements.

2.2 Data Source

Publicly available high-coverage (30x) whole-genome sequencing data from the 1000 Genomes Project (NYGC high-coverage release, GRCh38), accessed via the International Genome Sample Resource (IGSR) [6]. The dataset is explicitly consented for unrestricted public use. Five South Asian populations were included:

Code	Population	N
PJL	Punjabi in Lahore, Pakistan	146
BEB	Bengali in Bangladesh	131
STU	Sri Lankan Tamil in the UK	114
ITU	Indian Telugu in the UK	107
GIH	Gujarati Indian in Houston, TX	103
Total		601

2.3 Variant Extraction

PGx-relevant positions (43 rsIDs) were extracted from multi-sample 1000 Genomes VCFs using bcftools view with a curated BED file of GRCh38 coordinates. The resulting VCFs were split into 601 single-sample files for independent processing by each tool.

2.4 Gene Panel and Tier Classification

GenomeGuard defines 16 Tier 1 pharmacogenes (ABCG2, CYP1A2, CYP2B6, CYP2C19, CYP2C8, CYP2C9, CYP3A4, CYP3A5, DPYD, IFNL3, NAT2, NUDT15, SLCO1B1, TPMT, UGT1A1, VKORC1) that can be reliably interpreted from germline VCFs using rsID-based calling. Tier 2 genes (CYP2D6, HLA-A, HLA-B, G6PD) require specialist callers and are excluded from VCF-based concordance analysis but are supported by GenomeGuard's phenotype inference module when provided with orthogonal allele calls.

Of the 16 Tier 1 genes, only 13 produced callable results in both tools and were therefore eligible for head-to-head concordance comparison. Three genes were excluded from comparison for the following reasons:

- CYP1A2: PharmCAT's allele-definition files do not currently include CYP1A2 star-allele definitions for VCF-based calling.
- CYP2C8: PharmCAT does not produce CYP2C8 calls from the variant positions present in the input VCFs.
- NAT2: PharmCAT's Named Allele Matcher does not currently support NAT2 star-allele calling from short-read VCF data. GenomeGuard calls NAT2 using an rsID-based acetylator-status heuristic, but without a comparator result, concordance cannot be assessed.

These exclusions represent comparator capability gaps in PharmCAT's current release, not limitations of GenomeGuard. GenomeGuard produces calls for all 16 Tier 1 genes; the three excluded genes simply lack a reference result to validate against.

2.5 Pipeline Execution

GenomeGuard pipeline: VCF input is parsed by parser.py, which extracts variant records with rsID annotations. The analyzer.py module performs rsID lookup against a curated GRCh38 coordinate map (RSID_GRCH38), assigns star-alleles via pgx_knowledgebase.py, constructs diplotypes using a coverage-aware build_diplotype() function, and infers phenotypes using CPIC activity-score logic. Genes with no variant coverage at their defining positions are reported as Indeterminate rather than assumed wild-type, matching PharmCAT's conservative behavior.

PharmCAT pipeline: Each single-sample VCF is processed through the official pgkb/pharmcat Docker image, which runs the VCF Preprocessor (bgzip, tabix, normalisation), Named Allele Matcher, Phenotyper, and Reporter sequentially. Output JSON files are parsed to extract per-gene diplotype and phenotype calls.

2.6 Concordance Metrics

Diplotype concordance: Exact string match after canonical ordering (numerically lower allele listed first) and nomenclature normalization. Normalization handles PharmCAT's verbose allele labels (e.g., 'Reference/Reference' mapped to '*1/*1', 'c.1627A>G (*5) (heterozygous)' mapped to '*1/*5') and HGVS notation variants.

Phenotype concordance: Match after synonym normalization (e.g., 'Extensive Metabolizer' treated as equivalent to 'Normal Metabolizer' per CPIC's 2019 terminology update). Genes where one tool does not produce a phenotype call (e.g., IFNL3 in PharmCAT, which outputs 'n/a') were excluded from the phenotype concordance denominator to avoid artificial deflation.

2.7 Mismatch Triage

Each disagreement was classified by root cause in priority order: (1) Allele-definition version mismatch - PharmVar star-allele definition updates between tool versions causing different allele assignments from identical genotype data; (2) Missing variant coverage - VCF lacked genotype data at one or more defining positions for the gene; (3) Notation divergence - Both tools recognize the same underlying variant but express it through different nomenclature systems (rsID-based vs. star-allele vs. HGVS); (4) Logic difference - Genuine disagreement in the calling algorithm requiring manual review.

2.8 Performance Measurement

Execution times were measured as wall-clock elapsed time for processing the full cohort of 601 samples on identical hardware. GenomeGuard was timed using Python's time.time() around the complete analysis loop (VCF parsing through phenotype inference for all samples). PharmCAT was timed from the first Docker container launch through completion of the last sample's Reporter output. Both measurements include all I/O overhead. The PharmCAT timing reported here (3199.4s) represents a fresh, non-cached execution; subsequent runs with cached preprocessed VCFs completed faster but are not reported as they do not represent the end-to-end clinical workflow.

3. Results

3.1 Per-Gene Concordance

Table 1 presents per-gene diplotype and phenotype concordance across all 601 samples for the 13 genes with callable results in both tools. Ten of 13 genes achieved perfect (100.0%) diplotype concordance. The two genes with sub-100% concordance (UGT1A1 at 99.3% and ABCG2 at 100.0% diplotype / 99.7% phenotype) are discussed in detail in Section 3.3.

Table 1. Per-gene concordance between GenomeGuard and PharmCAT.

Gene	N	Diplotype (%)	Phenotype (%)	Mismatches
ABCG2	601	100.0	99.67	2
CYP2B6	601	100.0	100.0	0
CYP2C19	601	100.0	100.0	0
CYP2C9	601	100.0	100.0	0
CYP3A4	601	100.0	100.0	0
CYP3A5	601	100.0	100.0	0
DPYD	601	100.0	100.0	0
IFNL3	601	100.0	N/A*	0
NUDT15	601	100.0	100.0	0
SLCO1B1	601	100.0	100.0	0
TPMT	601	100.0	100.0	0
UGT1A1	598	99.33	99.83	4
VKORC1	601	100.0	100.0	0

**N/A: PharmCAT does not produce a phenotype classification for IFNL3; it outputs only the diplotype (rs12979860 genotype). GenomeGuard classifies IFNL3 as Favorable/Unfavorable Response per CPIC guidance. This gene was excluded from the phenotype concordance denominator.*

3.2 Overall Concordance

Across 7,810 sample-gene comparisons (13 genes x ~601 samples), GenomeGuard achieved 99.95% diplotype concordance and 99.96% phenotype concordance. A total of 6 sample-level mismatches were observed. Of these, 4 affected diplotype calls (all UGT1A1) and 3 affected phenotype calls (1 UGT1A1, 2 ABCG2); one mismatch contributed to both counts. No logic-level disagreements (Category 4) were detected in any gene.

3.3 Detailed Mismatch Analysis

All 6 mismatches are enumerated in Table 2 with their triaged root causes. Two distinct patterns were observed:

UGT1A1 (4 mismatches): PharmCAT assigned *37 or *36 where GenomeGuard assigned *28 for the same samples. The *28, *36, and *37 alleles share the (TA)n repeat polymorphism in the UGT1A1 promoter (rs8175347) but differ in repeat count. These alleles are defined by different versions of the PharmVar allele-definition tables; the discrepancy reflects version skew between the two tools' bundled PharmVar releases rather than an algorithmic error. Notably, the phenotype consequence was concordant in 3 of 4 cases (both tools assigned the same metabolizer status), confirming clinical equivalence.

ABCG2 (2 mismatches): Both tools identified the homozygous rs2231142 variant but expressed it through different nomenclature systems. GenomeGuard reported 'rs2231142_CA/rs2231142_CA' (rsID-based notation) while PharmCAT reported 'rs2231142 variant (T)/rs2231142 variant (T)' (allele-label notation). The diplotype strings were completely reconciled during normalization (resulting in 100.0% diplotype concordance); however, the phenotype diverged (GenomeGuard: 'Decreased Function'; PharmCAT: 'Poor Function'). This reflects a notation and functional-status mapping difference for the Q141K variant rather than a star-allele version mismatch. Both classifications are clinically actionable and would trigger the same CPIC recommendation for rosuvastatin dose adjustment.

Table 2. Complete enumeration of all sample-level mismatches.

Sample	Pop	Gene	GG Diplotype	PC Diplotype	GG Pheno	PC Pheno	Triage
HG02648	PJL	UGT1A1	*28/*28	*28/*37	Poor Metabolizer	Poor Metabolizer	allele_definition_version
HG02694	PJL	UGT1A1	*1/*28	*1/*36	Intermediate Metabolizer	Normal Metabolizer	allele_definition_version
HG04185	BEB	ABCG2	rs2231142_CA/rs2231142_CA	rs2231142 variant (T)/rs2231142 variant (T)	Decreased Function	Poor Function	notation_divergence
HG04191	BEB	UGT1A1	*1/*28	*1/*37	Intermediate Metabolizer	Intermediate Metabolizer	allele_definition_version
HG04193	BEB	UGT1A1	*1/*28	*1/*37	Intermediate Metabolizer	Intermediate Metabolizer	allele_definition_version
HG04210	STU	ABCG2	rs2231142_CA/rs2231142_CA	rs2231142 variant (T)/rs2231142 variant (T)	Decreased Function	Poor Function	notation_divergence

3.4 Genes Excluded from Head-to-Head Comparison

Three Tier 1 genes (CYP1A2, CYP2C8, NAT2) were excluded from concordance analysis because PharmCAT did not produce callable results for these genes from the input VCFs. This represents a capability gap in PharmCAT's current allele-definition coverage, not a limitation of GenomeGuard. GenomeGuard successfully produced star-allele calls for all three genes across all 601 samples. However, without a comparator result, concordance cannot be assessed, and these genes are excluded from all reported concordance metrics.

NAT2 merits specific discussion: NAT2 acetylator status is clinically important for isoniazid dosing in tuberculosis treatment, which is highly relevant to Indian healthcare. GenomeGuard implements NAT2 calling via an rsID-based rapid/intermediate/slow acetylator heuristic. PharmCAT does not currently support NAT2 calling. Future validation of GenomeGuard's NAT2 calls should be performed against orthogonal genotyping data or alternative reference tools that support NAT2.

3.5 Population-Stratified Concordance

Table 3 presents diplotype concordance stratified by the five South Asian populations. Concordance was uniformly high across all populations, with no evidence of population-specific systematic bias.

Table 3. Diplotype concordance (%) by population and gene.

Gene	BEB	GIH	ITU	PJL	STU

ABCG2	100.0	100.0	100.0	100.0	100.0
CYP2B6	100.0	100.0	100.0	100.0	100.0
CYP2C19	100.0	100.0	100.0	100.0	100.0
CYP2C9	100.0	100.0	100.0	100.0	100.0
CYP3A4	100.0	100.0	100.0	100.0	100.0
CYP3A5	100.0	100.0	100.0	100.0	100.0
DPYD	100.0	100.0	100.0	100.0	100.0
IFNL3	100.0	100.0	100.0	100.0	100.0
NUDT15	100.0	100.0	100.0	100.0	100.0
SLCO1B1	100.0	100.0	100.0	100.0	100.0
TPMT	100.0	100.0	100.0	100.0	100.0
UGT1A1	98.46	100.0	100.0	98.63	100.0
VKORC1	100.0	100.0	100.0	100.0	100.0

3.6 Performance Benchmarking

Both tools were benchmarked on the same hardware processing the identical 601-sample cohort. All timings are measured wall-clock elapsed time from a single, non-cached execution run.

Table 4. Measured execution time comparison.

Metric	GenomeGuard	PharmCAT
Total cohort time (wall-clock)	2.3s	3199.4s
Time per sample	3.8 ms	5323 ms
Time per gene call	0.29 ms	253 ms
Throughput (samples/sec)	261	0.19
Speedup factor	1391x	1x (baseline)

Under the tested deployment configuration, GenomeGuard demonstrated substantially lower execution time. GenomeGuard processed all 601 samples in 2.3 seconds, yielding a throughput of 261 samples per second. PharmCAT required 3199.4 seconds (53.3 minutes) for the same cohort, yielding 0.19 samples per second -- a 1391x difference in measured wall-clock time. The difference is attributable to GenomeGuard's native Python execution versus PharmCAT's Docker container orchestration overhead (container startup, JVM initialization, file-system mount overhead per sample). It is important to note that this comparison reflects the end-to-end deployment configurations of both tools; PharmCAT's core Java engine may perform differently when invoked natively without Docker.

4. Discussion

This study demonstrates that GenomeGuard produces pharmacogenomic calls that are clinically equivalent to PharmCAT across the 13 genes amenable to head-to-head comparison. The 99.95% diplotype concordance and 99.96% phenotype concordance, with zero logic-level disagreements, establish GenomeGuard as a reliable alternative for VCF-based PGx interpretation.

The six observed mismatches fall into two well-understood categories. The four UGT1A1 discrepancies reflect PharmVar allele-definition version skew: the *28, *36, and *37 alleles are distinguished by TA-repeat count at the promoter, and the specific allele assigned depends on which version of the definition table each tool bundles. The two ABCG2 discrepancies reflect a nomenclature and functional-status mapping difference for the Q141K (rs2231142) variant. Neither category represents a clinically significant divergence, as the downstream CPIC drug recommendations would be identical in all cases.

Under the tested deployment configuration, GenomeGuard demonstrated substantially lower execution time than PharmCAT. While this comparison reflects end-to-end deployment overhead (including Docker container orchestration for PharmCAT) rather than a pure algorithmic benchmark, the practical implication is significant: GenomeGuard's sub-second per-cohort execution enables real-time, point-of-care PGx interpretation without requiring server infrastructure or Docker environments. This is particularly relevant for deployment in Indian clinical settings where computational resources may be limited and where PGx results must be available within the clinical encounter window.

Beyond performance, GenomeGuard and PharmCAT serve fundamentally different design objectives. PharmCAT is primarily designed as a command-line annotation engine for researchers and bioinformaticians, producing structured JSON outputs suitable for downstream computational analysis. GenomeGuard, by contrast, focuses on clinical deployment through physician-facing reporting, medication safety screening, and healthcare workflow integration. Its analysis engine feeds directly into a drug interaction safety matrix, patient-facing reports, and family-level PGx dashboards designed for use by prescribing clinicians rather than bioinformatics specialists. This clinical orientation motivates the lightweight, dependency-free architecture: the tool must operate reliably in hospital IT environments where Docker installation, Java runtime management, and command-line access may not be available.

Three Tier 1 genes (CYP1A2, CYP2C8, NAT2) were excluded from the concordance comparison because PharmCAT's current release does not produce callable results for these genes. This is a capability gap in the comparator tool, not in GenomeGuard. NAT2 acetylator status is of particular clinical importance in Indian populations due to the high tuberculosis burden and isoniazid prescribing frequency. GenomeGuard's NAT2 calling should be validated in future work against orthogonal reference data or specialized NAT2 genotyping assays.

Tier 2 genes (CYP2D6, HLA-A, HLA-B, G6PD) were excluded by design from both tools' analysis. These genes require copy-number variation analysis, structural variant detection, or long-read phasing that cannot be reliably performed from short-read VCF data alone. This is an industry-wide limitation acknowledged by CPIC and PharmGKB, not specific to either tool.

4.1 Limitations

- This validation is limited to South Asian populations from the 1000 Genomes Project. Allele frequency distributions and rare variants in other ancestral groups may yield different concordance profiles. Multi-ethnic validation is planned.

- The 30x WGS data used here provides comprehensive coverage of PGx positions. Performance on lower-coverage data (e.g., clinical exome panels or targeted PGx panels) has not been assessed.

- Three Tier 1 genes could not be compared due to PharmCAT capability gaps. Validation of GenomeGuard's CYP1A2, CYP2C8, and NAT2 calls against alternative reference tools or genotyping assays remains necessary. As a preliminary check, we executed a Supplementary Internal Consistency Check against synthetic CPIC truth tables for the 3 omitted Tier 1 genes and 4 Tier 2 specialist genes. While GenomeGuard passed these limited unit tests (verifying correct transcription of CPIC logic), this does not replace the need for independent clinical validation.

- The ABCG2 functional-status classification ('Decreased Function' vs. 'Poor Function' for homozygous Q141K carriers) represents a genuine interpretive difference between the two tools that warrants further investigation against clinical outcome data.

5. Conclusions

GenomeGuard achieves 99.95% diplotype concordance and 99.96% phenotype concordance with PharmCAT across 13 pharmacogenes in a 601-sample South Asian validation cohort, with zero logic-level disagreements. Under the tested deployment configuration, GenomeGuard demonstrated substantially lower execution time (1391x difference), enabling real-time pharmacogenomic interpretation without Docker or cloud infrastructure. Combined with its physician-facing reporting and medication safety screening capabilities, these results support GenomeGuard's deployment as a clinically oriented PGx interpretation tool for South Asian healthcare settings.

6. Data Availability

All input data were derived from the publicly available 1000 Genomes Project (IGSR, https://www.internationalgenome.org/). The GenomeGuard source code, validation pipeline, and all intermediate results (concordance summaries, mismatch files, population breakdowns) are available at GenomeGuard.tech. PharmCAT was accessed via the official Docker image (pgkb/pharmcat:2.15.4).

Competing Interests

A.Y. is the developer of GenomeGuard and founder of GenomeGuard.tech. PharmCAT is an open-source tool developed by PharmGKB at Stanford University; the authors have no affiliation with PharmGKB or Stanford University.

Author Contributions

A.Y. conceived and designed the study, developed the GenomeGuard analysis engine, designed and implemented the validation pipeline, performed all computational analyses, and wrote the manuscript.

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Abbreviations

Abbreviation	Definition
CPIC	Clinical Pharmacogenetics Implementation Consortium
GRCh38	Genome Reference Consortium Human Build 38
IGSR	International Genome Sample Resource
PGx	Pharmacogenomics
rsID	Reference SNP Identifier
VCF	Variant Call Format
WGS	Whole-Genome Sequencing

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